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1. Introduction

This document describes a GitHub App that reviews pull requests using AWS Bedrock and Lambda. The authored functional architecture remains stable while deployment/runtime/code evidence is inferred from infrastructure manifests, source files, and verification artifacts.

1.1. Scope and Assumptions

- `Functional Groups` and `Functional Units` are the stable authored design backbone.
- `Runtime elements` and `code elements` are inferred from available IaC, deployment, and source artifacts.
- Inferred coverage is evidence-driven and may be partial when source metadata is missing.
- `Requirements` and reference indexes are provided for traceability, not as primary narrative flow.

2. How To Read This Document

This document is intentionally split into two layers:

1. Concept layer: functional and view-specific architecture narratives for human understanding.
2. Evidence layer: generated requirement traceability and inferred reference indexes.

Read in this order:

1. Architecture Intent View: stable authored architecture intent and ownership boundaries.
2. Communication / Deployment / Security Views: viewpoint-specific operational behavior.
3. Traceability View: requirement mapping, evidence coverage, and confidence.
4. Reference Appendix: supporting reference IDs and inferred-source links.

3. Document Health Snapshot

View	Authored Status	Authored Status Explanation	Units in Scope	Direct Evidence Coverage	Gap Count	Heuristic Confidence	Heuristic Basis
Architecture Intent View (`VIEW-ARCHITECTURE-INTENT`)	stable	Authored capability boundaries and ownership responsibilities are stable for this release.	7	7	0	high	Defaults authored units to covered; use authoredStatus as the primary publication signal.
Communication View (`VIEW-COMMUNICATION`)	in-review	Communication sequencing is defined, while protocol-level metadata is still being expanded.	7	3	4	low	Counts a unit as covered when direct inferred runtime evidence exists.
Deployment View (`VIEW-DEPLOYMENT`)	in-review	Deployment and runtime mapping are present; observability and ownership evidence are still evolving.	7	3	4	low	Counts a unit as covered when direct inferred runtime evidence exists.

View	Authored Status	Authored Status Explanation	Units in Scope	Direct Evidence Coverage	Gap Count	Heuristic Confidence	Heuristic Basis
Security View (`VIEW-SECURITY`)	draft	Core threats are modeled; control-depth and detection mappings are still being expanded.	7	7	2	high	Counts a unit as covered when explicit authored attack-vector mapping exists.
Traceability View (`VIEW-TRACEABILITY`)	in-review	Requirement mappings exist; evidence completeness and consistency are still being hardened.	7	7	4	high	Counts a unit as covered when direct inferred runtime or code evidence exists.

4. Terms and Definitions

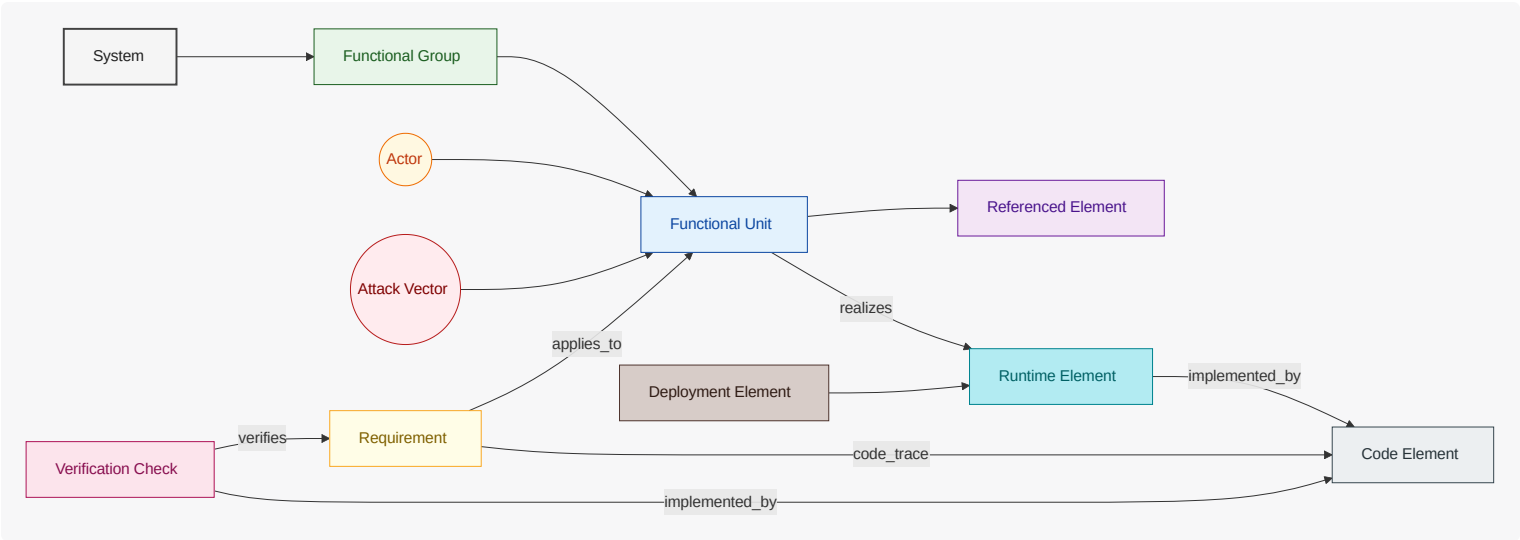
Term	Definition
`actor` (EM-ACTOR)	A person or operational role that interacts with functional units.
`ai review is enabled` (FEAT-AI-REVIEW)	Feature flag enabling model-assisted review. Aliases:ai review enabled
`attack vector` (EM-ATTACK-VECTOR)	A technical misuse or attack path that targets functional, referenced, or runtime elements.
`audit logging is enabled` (FEAT-AUDIT-LOGGING)	Feature flag enabling audit metadata persistence. Aliases:audit enabled
`authored mapping` (EM-AUTHORED-MAPPING)	An explicit relationship declared in architecture inputs between authored or referenced elements.
`aws sdk for go` (REF-AWS-SDK-GO)	SDK dependency used for AWS runtime integrations. Aliases:aws sdk
`aws secrets manager` (REF-AWS-SECRETS-MANAGER)	Managed secret store for application credentials and config values. Aliases:secrets manager
`bedrock runtime` (REF-AWS-BEDROCK-RUNTIME)	External Bedrock runtime endpoint used for model inference. Aliases:bedrock api
`bedrock runtime is unavailable` (EVT-BEDROCK-UNAVAILABLE)	Event raised when Bedrock inference path is degraded. Aliases:bedrock unavailable
`check run summary` (DATA-CHECKRUN-SUMMARY)	Aggregated review result posted as a GitHub check run.
`code element` (EM-CODE-ELEMENT)	An inferred code structure or ownership element discovered from source trees and build metadata.
`developer` (ACT-DEVELOPER)	Engineer opening and updating pull requests. Aliases:pr author
`diff context` (DATA-DIFF-CONTEXT)	Pull request file changes and metadata prepared for analysis.
`diff context is prepared` (EVT-CONTEXT-PREPARED)	Event indicating context assembly completed and is ready for analysis. Aliases:context prepared
`functional group` (EM-FUNCTIONAL-GROUP)	A major authored capability area that groups related functional units.

Term	Definition
functional unit (EM-FUNCTIONAL-UNIT)	An authored working unit inside a functional group that owns specific behavior.
github app api (REF-GITHUB-APP-API)	External GitHub API used for pull request metadata and review publication. Aliases:github api
github rest sdk (REF-GITHUB-REST-SDK)	SDK dependency used for GitHub App REST interactions. Aliases:github sdk
github webhook ingress (FU-GITHUB-WEBHOOK-INGRESS)	Validates webhook signatures and routes trusted pull request events. Aliases:webhook ingress
inferred mapping (EM-INFERRED-MAPPING)	A discovered relationship that links inferred runtime/code elements upward to authored design.
lambda runtime operations (FU-LAMBDA-RUNTIME-OPERATIONS)	Manages runtime deployment and operational controls. Aliases:lambda ops
normal mode is active (MODE-NORMAL)	Default operating mode using AI plus deterministic checks. Aliases:normal mode
platform operations (FG-PLATFORM-OPERATIONS)	Functional group for deployment lifecycle and secret/config controls. Aliases:runtime operations
platform operator (ACT-PLATFORM-OPERATOR)	Owner of deployment/runtime operations and reliability controls. Aliases:platform ops
policy checks (FU-POLICY-CHECKS)	Evaluates deterministic repository policy checks. Aliases:deterministic checks
policy checks are enabled (FEAT-POLICY-CHECKS)	Feature flag enabling deterministic policy checks. Aliases:policy checks enabled
policy ruleset (REF-POLICY-RULESET)	Deterministic review policy bundle used during analysis. Aliases:policy bundle
pr context assembly (FU-PR-CONTEXT-ASSEMBLY)	Fetches changed files and prepares bounded review context payloads. Aliases:context assembly
pr integration (FG-PR-INTEGRATION)	Functional group for webhook ingress and pull request context intake. Aliases:github integration
prompt injection in diff (AV-PROMPT-INJECTION-IN-DIFF)	Malicious prompt-like content embedded in pull request code changes. Aliases:prompt injection
pull request opened event is received (EVT-PR-OPENED)	Event raised when a new pull request is opened. Aliases:pr opened
pull request review is active (STATE-PR-REVIEW-ACTIVE)	State where review processing is currently active for a pull request. Aliases:review active
pull request spam abuse (AV-PR-SPAM-ABUSE)	High-volume event abuse causing review capacity exhaustion. Aliases:pr spam
pull request synchronize event is received (EVT-PR-SYNCHRONIZE)	Event raised when new commits are pushed to an open pull request. Aliases:pr synchronize
redacted audit metadata (DATA-REDACTED-AUDIT-METADATA)	Sanitized prompt and finding metadata retained for audit.
referenced element (EM-REFERENCED-ELEMENT)	An architecture-relevant external, platform, or third-party dependency represented by role.

Term	Definition
`repository maintainer` (ACT-REPOSITORY-MAINTAINER)	Owner of merge decisions and repository quality gates. Aliases:maintainer
`requirement` (EM-REQUIREMENT)	A structured requirement statement that defines expected system behavior and maps to functional ownership.
`result metadata` (DATA-RESULT-METADATA)	Metadata describing review output characteristics for audit and reporting.
`review findings are generated` (EVT-FINDINGS-GENERATED)	Event indicating analysis output is ready for publication. Aliases:findings generated
`review findings are published` (EVT-FINDINGS-PUBLISHED)	Event indicating review output has been published to GitHub checks/comments. Aliases:findings published
`review intelligence` (FG-REVIEW-INTELLIGENCE)	Functional group for AI and deterministic review analysis and publication. Aliases:pr review system
`review orchestration` (FU-REVIEW-ORCHESTRATION)	Coordinates Bedrock analysis with deterministic checks and fallback logic. Aliases:orchestration
`review publication` (FU-REVIEW-PUBLICATION)	Publishes review findings to GitHub checks and comments. Aliases:publication
`review request rate limit is exceeded` (EVT-RATE-LIMIT-EXCEEDED)	Event raised when PR review rate guardrails are hit. Aliases:rate limit exceeded
`runtime element` (EM-RUNTIME-ELEMENT)	An inferred runtime realization element discovered from infrastructure and deployment sources.
`secret leakage in review output` (AV-SECRET-LEAKAGE-IN-REVIEW)	Unintended credential disclosure in generated review findings. Aliases:secret leakage
`secrets and configuration` (FU-SECRETS-CONFIGURATION)	Manages credentials and secure runtime configuration. Aliases:secret config
`security engineer` (ACT-SECURITY-ENGINEER)	Owner of security policy checks and finding triage. Aliases:security reviewer
`spoofed github webhook` (AV-SPOOFED-GITHUB-WEBHOOK)	Forged webhook request attempting unauthorized review execution. Aliases:webhook spoofing
`upward linking` (EM-UPWARD-LINKING)	Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs.
`verification check` (EM-VERIFICATION-CHECK)	An inferred or authored verification artifact that validates requirement behavior with test evidence.
`webhook signature` (DATA-WEBHOOK-SIGNATURE)	Signature used to verify webhook authenticity.
`webhook signature is valid` (COND-WEBHOOK-SIGNATURE-VALID)	Condition indicating webhook HMAC verification succeeded. Aliases:signature valid

4.1. Diagram Color and Shape Legend

This legend applies to all Mermaid diagrams in this document.



5. Architecture Intent View

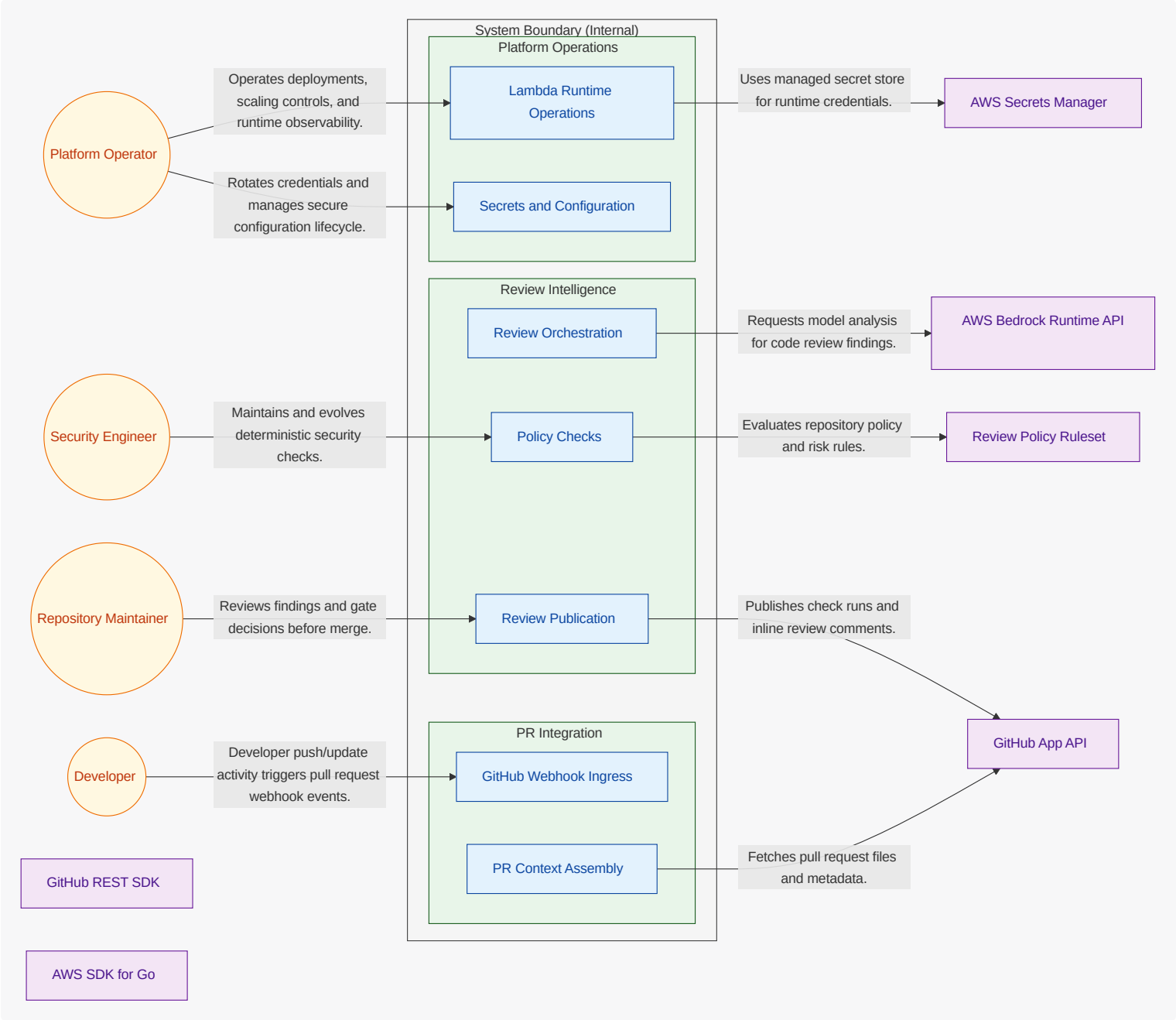
Show the stable authored architecture intent: capability areas, unit ownership boundaries, and intentional responsibility split.

5.1. What This View Answers

- What are the main capability areas?
- What are the stable units of responsibility and what does each unit own?
- How are responsibilities intentionally split?

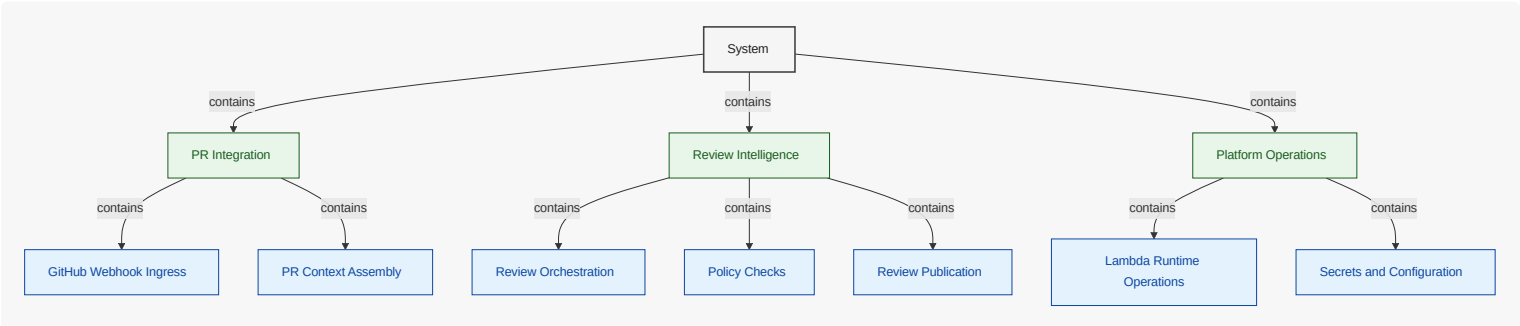
5.2. System Boundary Diagram

What this diagram shows: explicit internal vs external scope. Functional units are inside the `System Boundary (Internal)` container, while actors and referenced dependencies are shown outside the boundary.



5.3. System Decomposition Diagram

What this diagram shows: stable capability decomposition from functional groups into functional units.



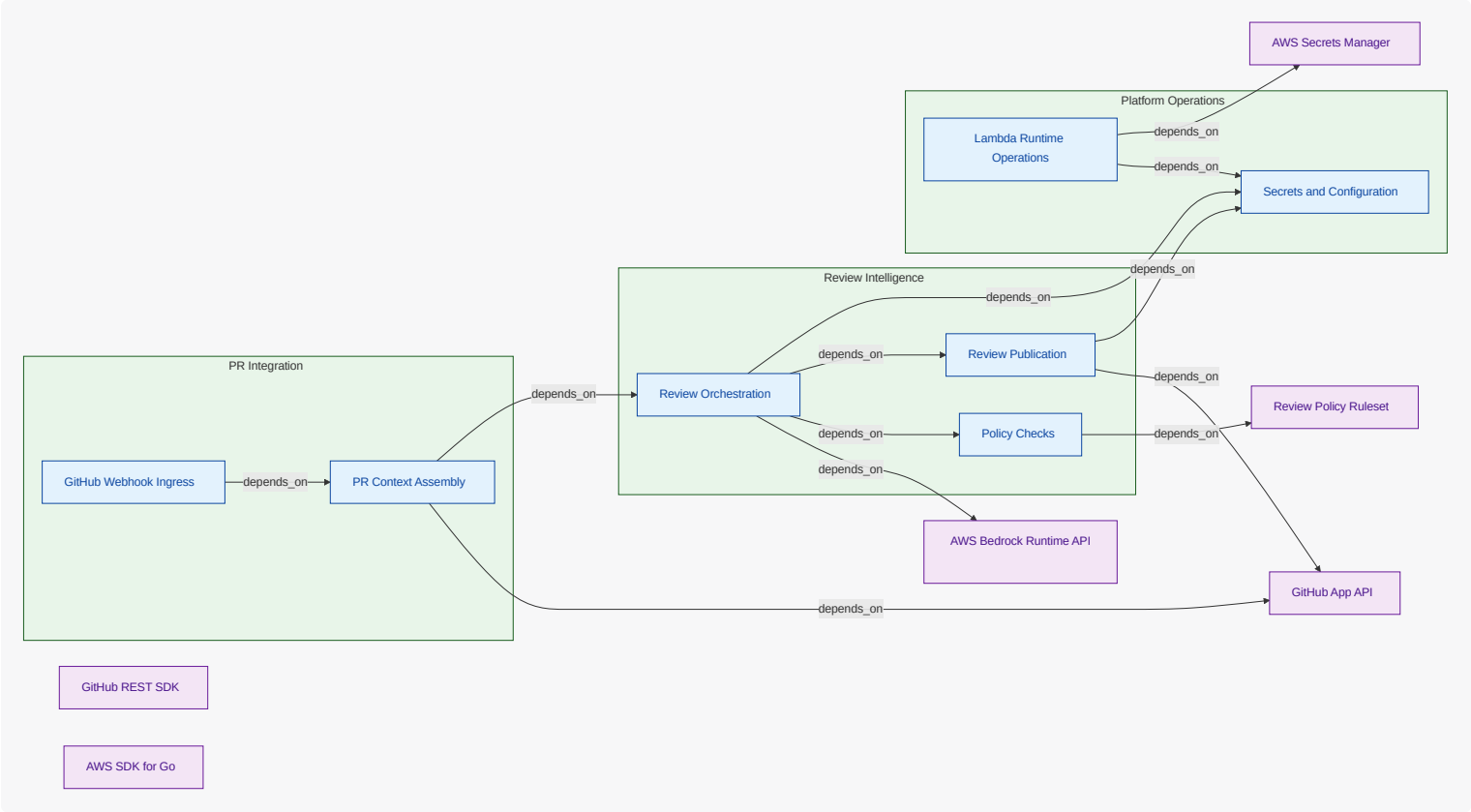
5.4. Functional Group Manhattan Matrix

What this diagram shows: a table-like matrix where functional groups are the bottom row (columns) and functional units are arranged in rows above their owning group.

	Policy Checks	
GitHub Webhook Ingress	Review Orchestration	Lambda Runtime Operations
PR Context Assembly	Review Publication	Secrets and Configuration
PR Integration	Review Intelligence	Platform Operations

5.5. Functional Unit Dependency Diagram

What this diagram shows: dependency paths between functional units (grouped in Functional Group containers) and key referenced elements. Arrow meaning: `A -→|depends_on| B` means unit `A` requires behavior or data from `B` to fulfill its responsibility.



5.6. Functional Groups and Units (View Scoped)

5.6.1. PR Integration (FG-PR-INTEGRATION)

Ingress and `context assembly` for GitHub pull request events.

`PR Integration` owns webhook trust boundaries and pull request context intake. It converts incoming GitHub events into normalized review work with clear boundaries.

GitHub Webhook Ingress (FU-GITHUB-WEBHOOK-INGRESS)

Verifies `webhook signatures`, validates event type, and starts review workflow routing.

Owned Responsibility

functional responsibility for `github webhook ingress`; includes decision and `orchestration` flow to `PR Context Assembly`

Key Collaborators

`PR Context Assembly`

Linked Requirements

`REQ-PRR-001`, `REQ-PRR-008`

PR Context Assembly (FU-PR-CONTEXT-ASSEMBLY)

Fetches changed files and diff metadata and prepares bounded review context payloads.

Owned Responsibility

functional responsibility for `pr context assembly`; includes decision and `orchestration` flow to `GitHub App API` and decision and `orchestration` flow to `Review Orchestration`

Key Collaborators

`GitHub App API`, `Review Orchestration`

Linked Requirements

`REQ-PRR-002`

5.6.2. Review Intelligence (FG-REVIEW-INTELLIGENCE)

AI and deterministic review analysis with result `publication`.

`Review Intelligence` owns generated findings, `deterministic checks`, and final `publication` behavior. It combines Bedrock reasoning with predictable `policy checks` and emits auditable review outputs.

Policy Checks (FU-POLICY-CHECKS)

Runs `deterministic checks` for secrets, risky IaC changes, and repository policy constraints.

Owned Responsibility

functional responsibility for `policy checks`; includes decision and `orchestration` flow to `Review Policy Ruleset`

Key Collaborators

`Review Policy Ruleset`

Linked Requirements

`REQ-PRR-004`, `REQ-PRR-006`

Review Orchestration (FU-REVIEW-ORCHESTRATION)

Calls Bedrock models, merges AI and deterministic findings, and handles graceful degradation paths.

Owned Responsibility

functional responsibility for `review orchestration`; includes decision and `orchestration` flow to `AWS Bedrock Runtime API` and decision and `orchestration` flow to `Policy Checks`

Key Collaborators

`AWS Bedrock Runtime API`, `Policy Checks`, `Review Publication`, `Secrets and Configuration`

Linked Requirements

`REQ-PRR-003`, `REQ-PRR-006`

Review Publication (FU-REVIEW-PUBLICATION)

Publishes check-run summaries and inline comments back to GitHub pull requests.

Owned Responsibility

functional responsibility for `review publication`; includes decision and `orchestration` flow to `GitHub App API` and decision and `orchestration` flow to `Secrets and Configuration`

Key Collaborators

`GitHub App API`, `Secrets and Configuration`

Linked Requirements

`REQ-PRR-005`, `REQ-PRR-006`, `REQ-PRR-007`, `REQ-PRR-008`

5.6.3. Platform Operations (FG-PLATFORM-OPERATIONS)

Lambda deployment lifecycle, `runtime operations`, and secrets/configuration.

`Platform Operations` owns safe deployment, runtime controls, and secret/config lifecycle for the app. It enables dependable review operation without owning business-level review policy semantics.

Lambda Runtime Operations (FU-LAMBDA-RUNTIME-OPERATIONS)

Manages Lambda packaging/deployment lifecycle and operational health guardrails.

Owned Responsibility

functional responsibility for `lambda runtime operations`; includes decision and `orchestration` flow to `AWS Secrets Manager` and decision and `orchestration` flow to `Secrets and Configuration`

Key Collaborators

`AWS Secrets Manager`, `Secrets and Configuration`

Linked Requirements

no explicit authored `requirement` trigger

Secrets and Configuration (FU-SECRETS-CONFIGURATION)

Manages GitHub App credentials, Bedrock config, and redaction-safe runtime configuration.

Owned Responsibility

functional responsibility for `secrets and configuration`

Key Collaborators

none

Linked Requirements

`REQ-PRR-007`

6. Communication View

Show who communicates with whom across key request paths, including interface edges and dependency handoffs.

6.1. What This View Answers

- Who talks to whom and over which interfaces?
- Which edges are synchronous, asynchronous, event-driven, or callback-oriented?
- Which communication paths are critical to the main request flow?

6.2. Coverage Gaps

Coverage summary: 3/7 units have direct evidence coverage in this view.

- `Lambda Runtime Operations` has no direct inferred runtime/deployment evidence yet.
- `PR Context Assembly` has no direct inferred runtime/deployment evidence yet.
- `Policy Checks` has no direct inferred runtime/deployment evidence yet.
- `Secrets and Configuration` has no direct inferred runtime/deployment evidence yet.

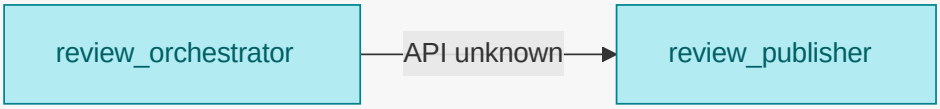
6.3. Recommended Next Evidence Additions

- Add explicit communication metadata (protocol, sync/async/event/callback) to `authored mappings` where currently implicit.
- Expose service protocol/port metadata in Helm or manifest values to enrich interface-edge inference.
- Capture missing communicating runtime units by adding discoverable workload artifacts under scanned inference roots.

6.4. Communication Path Graph

What this diagram shows: inferred unit-to-unit request/call flow, including available interface/port labels.

> **NOTE** This graph is inferred from available deployment/runtime artifacts and may be partial. Missing edges or interface details indicate missing derived communication evidence, not necessarily missing behavior.



6.5. Functional Groups and Units (View Scoped)

6.5.1. PR Integration (FG-PR-INTEGRATION)

`PR Integration` receives webhook events and forwards normalized review context to `orchestration` with explicit handoff semantics.

GitHub Webhook Ingress (FU-GITHUB-WEBHOOK-INGRESS)

Receives webhook payloads, validates them, and routes authorized events to `context assembly`.

Inbound Interfaces

Interface	Description
`Developer`	`Developer` push/update activity triggers pull request webhook events.

Outbound Interfaces

Interface	Description
`PR Context Assembly`	Forwards authenticated PR events for `context assembly`.

Communication Evidence

runtime: `webhook_ingress` | code modules: `webhook_ingress`

PR Context Assembly (FU-PR-CONTEXT-ASSEMBLY)

Calls GitHub APIs and forwards normalized context to `orchestration`.

Inbound Interfaces

Interface	Description
`GitHub Webhook Ingress`	Forwards authenticated PR events for `context assembly`.

Outbound Interfaces

Interface	Description
`GitHub App API`	Fetches pull request files and metadata.
`Review Orchestration`	Sends normalized review context.

Communication Evidence

code modules: `pr_context_assembly`

6.5.2. Review Intelligence (FG-REVIEW-INTELLIGENCE)

`Review Intelligence` coordinates Bedrock calls, policy evaluation, and `publication` sequencing for deterministic outcomes.

Policy Checks (FU-POLICY-CHECKS)

Exchanges policy findings with `orchestration` for final result composition.

Inbound Interfaces

Interface	Description
`Review Orchestration`	Merges deterministic policy findings with model output.
`Security Engineer`	Maintains and evolves deterministic security checks.

Outbound Interfaces

Interface	Description
`Review Policy Ruleset`	Evaluates repository policy and risk rules.

Communication Evidence

code modules: `policy_checks`

Review Orchestration (FU-REVIEW-ORCHESTRATION)

Integrates context input, Bedrock calls, and `publication` output in a deterministic sequence.

Inbound Interfaces

Interface	Description
`PR Context Assembly`	Sends normalized review context.

Outbound Interfaces

Interface	Description
`AWS Bedrock Runtime API`	Requests model analysis for code review findings.
`Policy Checks`	Merges deterministic policy findings with model output.
`Review Publication`	Submits final findings for `publication`.
`Secrets and Configuration`	Retrieves model configuration and safety settings.

Communication Evidence

runtime: `review_orchestrator` | code modules: `review_orchestrator`

Review Publication (FU-REVIEW-PUBLICATION)

Posts summaries and inline annotations through `GitHub App API`s.

Inbound Interfaces

Interface	Description
`Repository Maintainer`	Reviews findings and gate decisions before merge.
`Review Orchestration`	Submits final findings for `publication`.

Outbound Interfaces

Interface	Description
`GitHub App API`	Publishes check runs and inline review comments.
`Secrets and Configuration`	Retrieves app credentials and redaction rules.

Communication Evidence

runtime: `review_publisher` | code modules: `review_publication`

6.5.3. Platform Operations (FG-PLATFORM-OPERATIONS)

`Platform Operations` coordinates runtime controls with app units via deployment and secret/config integration interfaces.

Lambda Runtime Operations (FU-LAMBDA-RUNTIME-OPERATIONS)

Operates deployment and runtime controls that indirectly shape request-path reliability.

Inbound Interfaces

Interface	Description
`Platform Operator`	Operates deployments, scaling controls, and runtime observability.

Outbound Interfaces

Interface	Description
`AWS Secrets Manager`	Uses managed secret store for runtime credentials.
`Secrets and Configuration`	Requires secure runtime configuration and credential access.

Communication Evidence

code modules: `lambda_runtime_ops`

Secrets and Configuration (FU-SECRETS-CONFIGURATION)

Provides secure config retrieval contracts to `orchestration` and `publication` units.

Inbound Interfaces

Interface	Description
`Lambda Runtime Operations`	Requires secure runtime configuration and credential access.
`Platform Operator`	Rotates credentials and manages secure configuration lifecycle.
`Review Orchestration`	Retrieves model configuration and safety settings.
`Review Publication`	Retrieves app credentials and redaction rules.

Outbound Interfaces

none

Communication Evidence

code modules: `secrets_configuration`

7. Deployment View

Show inferred deployment artifacts and ownership mapping to authored units. This view emphasizes deployment relationships and `platform operations`.

7.1. What This View Answers

- How does authored intent become deployed runtime?
- What control-plane objects manage deployable units?
- Where are platform ownership boundaries and missing ownership signals?

7.2. Coverage Gaps

Coverage summary: 3/7 units have direct evidence coverage in this view.

- `Lambda Runtime Operations` has no direct inferred runtime/deployment evidence yet.
- `PR Context Assembly` has no direct inferred runtime/deployment evidence yet.
- `Policy Checks` has no direct inferred runtime/deployment evidence yet.
- `Secrets and Configuration` has no direct inferred runtime/deployment evidence yet.

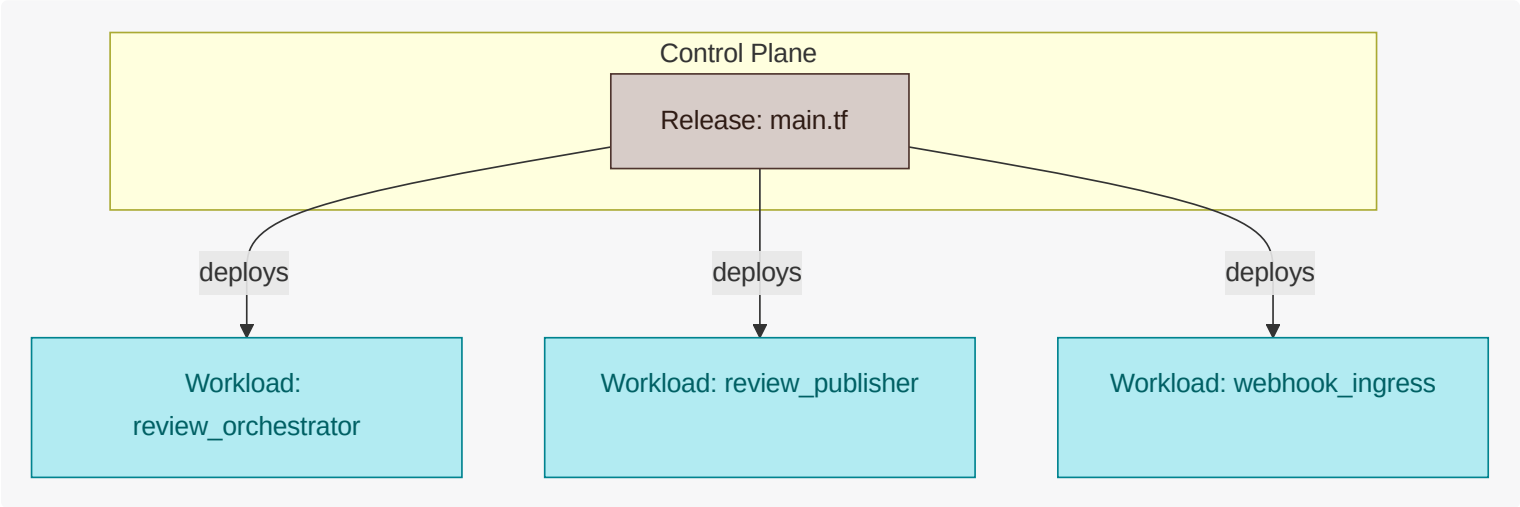
7.3. Recommended Next Evidence Additions

- Add owner metadata to Flux/Kustomization/Helm objects for clearer platform ownership mapping.
- Ensure deployment control artifacts (source, kustomization, release) are all included in scanned directories.
- Add deterministic naming conventions for release and namespace objects to improve cross-artifact linking.

7.4. Deployment Artifact Relationship Graph

What this diagram shows: how source/control-plane artifacts connect to releases, namespaces, workloads, and target cluster context.

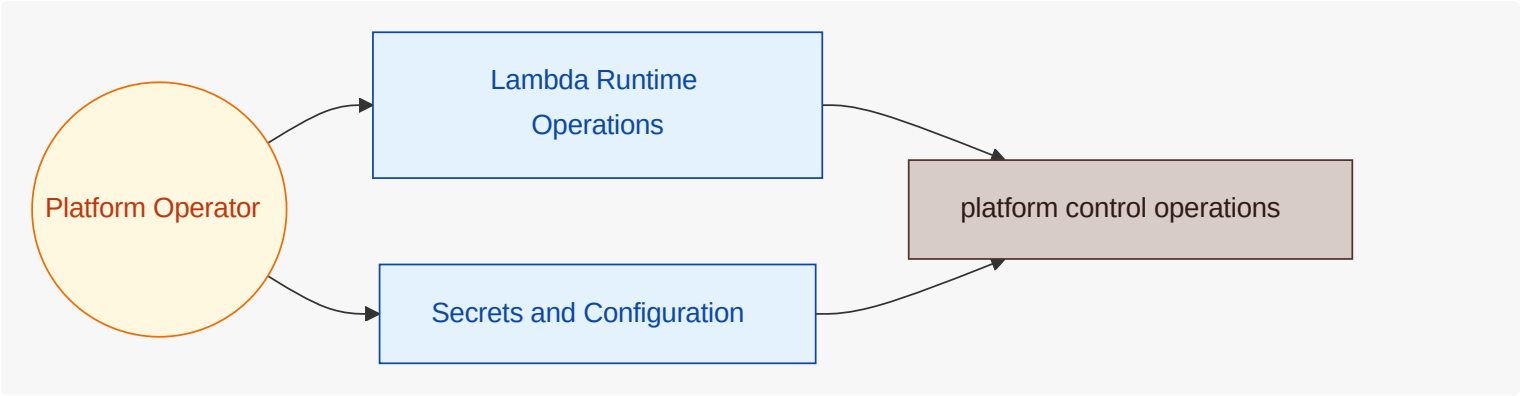
> **NOTE**
> This graph is inferred from available deployment artifacts (for example Terraform, Flux, Helm) and may be partial when source metadata is incomplete. Namespace and cluster boundaries are rendered as containers when target/placement relationships are inferred. Node labels include role hints (for example `Source`, `Kustomization`, `Release`, `Workload`, `ns/…`) to distinguish similarly named elements.



7.5. Platform Operations Graph

What this diagram shows: which platform actors operate which platform units and the control-plane targets they influence.

NOTE | Platform operations links are inferred from authored actor mappings plus discovered deployment control-plane artifacts.



7.6. Functional Groups and Units (View Scoped)

7.6.1. PR Integration (FG-PR-INTEGRATION)

`PR Integration` functions deploy independently to isolate `webhook ingress` from context-fetch implementation changes.

GitHub Webhook Ingress (FU-GITHUB-WEBHOOK-INGRESS)

GitHub Webhook Ingress Deployment

Deploys as an independently managed Lambda runtime boundary.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-PRR-001`, `REQ-PRR-008`
Deployment dependencies	`PR Context Assembly`
Inferred deployment evidence	runtime: `webhook_ingress`

PR Context Assembly (FU-PR-CONTEXT-ASSEMBLY)

PR Context Assembly Deployment

Deployed separately to allow context shaping changes without ingress disruption.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-PRR-002`
Deployment dependencies	`GitHub App API`, `Review Orchestration`
Inferred deployment evidence	code modules: `pr_context_assembly`

7.6.2. Review Intelligence (FG-REVIEW-INTELLIGENCE)

`Review Intelligence` units roll independently to preserve availability and reduce blast radius during model or rule updates.

Policy Checks (FU-POLICY-CHECKS)

Policy Checks Deployment

Deployed independently to evolve policy logic without model coupling.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-PRR-004`, `REQ-PRR-006`

Deployment Aspect	View-Scoped Detail
Deployment dependencies	`Review Policy Ruleset`
Inferred deployment evidence	code modules: `policy_checks`

Review Orchestration (FU-REVIEW-ORCHESTRATION)

Review Orchestration Deployment

Isolated deployment enables cautious model/config rollout and fallback control.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-PRR-003`, `REQ-PRR-006`
Deployment dependencies	`AWS Bedrock Runtime API`, `Policy Checks`, `Review Publication`, `Secrets and Configuration`
Inferred deployment evidence	runtime: `review_orchestrator`

Review Publication (FU-REVIEW-PUBLICATION)

Review Publication Deployment

Release isolation protects `publication` reliability during other unit rollouts.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-PRR-005`, `REQ-PRR-006`, `REQ-PRR-007`, `REQ-PRR-008`
Deployment dependencies	`GitHub App API`, `Secrets and Configuration`
Inferred deployment evidence	runtime: `review_publisher`

7.6.3. Platform Operations (FG-PLATFORM-OPERATIONS)

`Platform Operations` governs release `orchestration` and runtime placement to keep review capacity stable.

Lambda Runtime Operations (FU-LAMBDA-RUNTIME-OPERATIONS)

Lambda Runtime Operations Deployment

Governs source-to-release pipeline behavior and runtime placement boundaries.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	no explicit authored `requirement` trigger
Deployment dependencies	`AWS Secrets Manager`, `Secrets and Configuration`
Inferred deployment evidence	code modules: `lambda_runtime_ops`

Secrets and Configuration (FU-SECRETS-CONFIGURATION)

Secrets and Configuration Deployment

Managed as a shared operational boundary with controlled rotation and rollout flow.

Deployment Aspect	View-Scoped Detail
Deployment trigger context	`REQ-PRR-007`

Deployment Aspect	View-Scoped Detail
Deployment dependencies	none
Inferred deployment evidence	code modules: `secrets_configuration`

8. Security View

Show inferred exposure and dependency risk points aligned to unit boundaries, focused on attack paths and security evidence.

8.1. What This View Answers

- What are the main trust boundaries and attack paths?
- Which units are exposed and what security evidence exists?
- Where is the threat model incomplete?

8.2. Coverage Gaps

Coverage summary: 5/7 units have direct evidence coverage in this view.

- `Lambda Runtime Operations` has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.
- `Policy Checks` has no explicit authored attack-vector mapping yet; document technical and fraud/abuse threats even when mitigated.

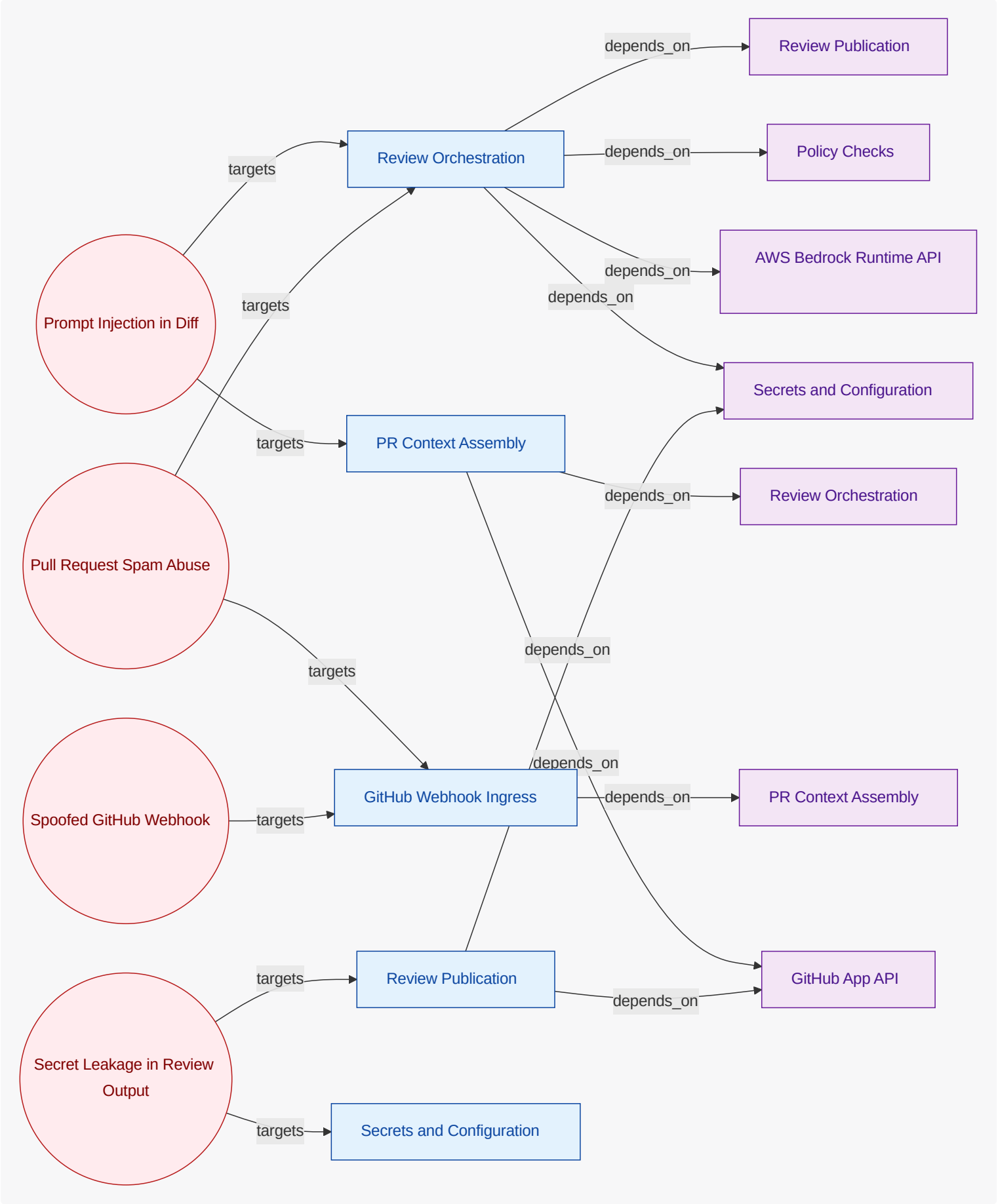
8.3. Recommended Next Evidence Additions

- Add explicit authored attack-vector mappings for units currently marked without threat linkage.
- Add security signal ownership metadata (logs/alerts/events) for unresolved observability evidence.
- Map additional security-relevant dependencies and exposures into `authored mappings` for traceable coverage.

8.4. Attack Path Diagram

What this diagram shows: authored attack vectors, their targeted units, and security-relevant dependency paths.

> **NOTE**
> Security relationships shown here are explicitly modeled or inferred from available evidence and may be incomplete. Absence of a path does not imply absence of risk. Threats that are currently mitigated (including fraud/abuse threats) should still be modeled as attack vectors for auditability and control-trace visibility.


Prompt Injection in Diff
Pull Request Spam Abuse
Spoofed GitHub Webhook
Secret Leakage in Review Output
Review Orchestration
PR Context Assembly
GitHub Webhook Ingress
Review Publication
Review Publication
Policy Checks
AWS Bedrock Runtime API
Secrets and Configuration
Review Orchestration
PR Context Assembly
GitHub App API
Secrets and Configuration
targets
targets
targets
targets
targets
targets
targets
depends_on
depends_on
depends_on
depends_on
depends_on
depends_on
depends_on
depends_on
depends_on

8.5. Functional Groups and Units (View Scoped)

8.5.1. PR Integration (FG-PR-INTEGRATION)

`PR Integration` protects signature verification and hostile-diff ingress paths to reduce spoofing and `prompt injection` risk.

GitHub Webhook Ingress (FU-GITHUB-WEBHOOK-INGRESS)

Enforces signature verification and rejects spoofed or malformed events.

Attack Surface

`PR Context Assembly`

Main Threats

`Pull Request Spam Abuse`, `Spoofed GitHub Webhook`

Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

Security Evidence Present

runtime: `webhook_ingress` | code modules: `webhook_ingress`

Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

PR Context Assembly (FU-PR-CONTEXT-ASSEMBLY)

Applies contextual sanitization to reduce `prompt injection` risk from diffs.

Attack Surface

`GitHub App API`, `Review Orchestration`

Main Threats

`Prompt Injection in Diff`

Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

Security Evidence Present

code modules: `pr_context_assembly`

Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

8.5.2. Review Intelligence (FG-REVIEW-INTELLIGENCE)

`Review Intelligence` emphasizes prompt hardening, output redaction, and secure `publication` boundaries.

Policy Checks (FU-POLICY-CHECKS)

Provides deterministic controls for secrets and risky infrastructure changes.

Attack Surface

`Review Policy Ruleset`

Main Threats

no explicit `attack vector` targeting this unit

Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

Security Evidence Present

code modules: `policy_checks`

Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

Review Orchestration (FU-REVIEW-ORCHESTRATION)

Enforces safe prompt construction and bounded output handling under adversarial input.

Attack Surface

`AWS Bedrock Runtime API`, `Policy Checks`, `Review Publication`, `Secrets and Configuration`

Main Threats

`Prompt Injection in Diff`, `Pull Request Spam Abuse`

Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

Security Evidence Present

runtime: `review_orchestrator` | code modules: `review_orchestrator`

Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

Review Publication (FU-REVIEW-PUBLICATION)

Applies redaction and policy-aware output filtering before publishing findings.

Attack Surface

`GitHub App API`, `Secrets and Configuration`

Main Threats

`Secret Leakage in Review Output`

Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

Security Evidence Present

runtime: `review_publisher` | code modules: `review_publication`

Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

8.5.3. Platform Operations (FG-PLATFORM-OPERATIONS)

`Platform Operations` secures credential paths and operational controls across Lambda deployment lifecycle.

Lambda Runtime Operations (FU-LAMBDA-RUNTIME-OPERATIONS)

Hardens operational control loops and protects deployment trust boundaries.

Attack Surface

`AWS Secrets Manager`, `Secrets and Configuration`

Main Threats

no explicit `attack vector` targeting this unit

Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

Security Evidence Present

code modules: `lambda_runtime_ops`

Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

Secrets and Configuration (FU-SECRETS-CONFIGURATION)

Enforces credential isolation, least-privilege access, and redaction-safe metadata handling.

Attack Surface

none

Main Threats

`Secret Leakage in Review Output`

Expected Controls

Define control expectations in authored security narratives and mapped dependencies.

Security Evidence Present

code modules: `secrets_configuration`

Security Evidence Missing

Close any unresolved ownership or unmapped threat-linkage gaps in Traceability View.

9. Traceability View

Show requirement-to-unit-to-evidence traceability, including coverage confidence and explicit evidence gaps.

9.1. What This View Answers

- Which `requirements` map to which units?
- Which authored units have runtime/code evidence and which do not?
- Where are confidence or completeness gaps in trace coverage?

9.2. Coverage Gaps

Coverage summary: 7/7 units have direct evidence coverage in this view.

- `Lambda Runtime Operations` has no direct inferred runtime evidence yet.
- `PR Context Assembly` has no direct inferred runtime evidence yet.
- `Policy Checks` has no direct inferred runtime evidence yet.
- `Secrets and Configuration` has no direct inferred runtime evidence yet.

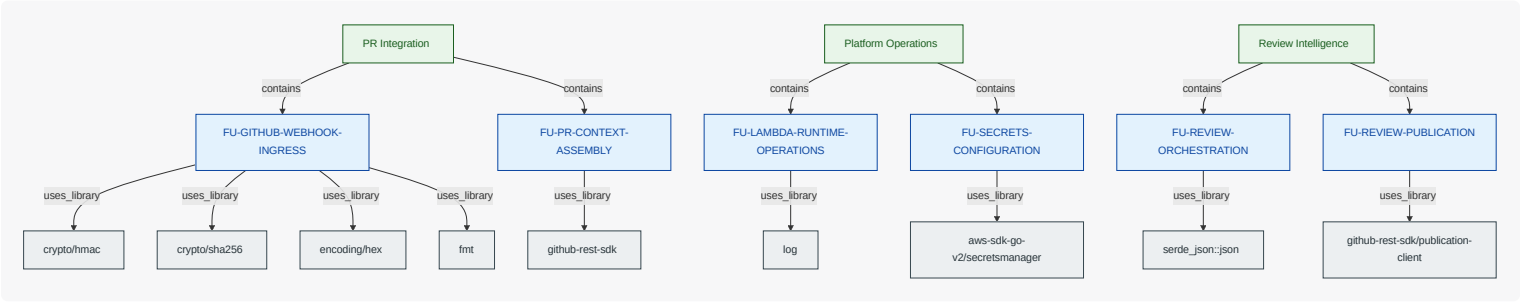
9.3. Recommended Next Evidence Additions

- Add explicit requirement-to-owner mappings where `requirement` scope is currently broad or ambiguous.
- Add ownership annotations for runtime and code artifacts where coverage is still unresolved.
- Prioritize closing gaps for units that are authored but lack both runtime and code evidence.

9.4. Traceability Evidence Graph

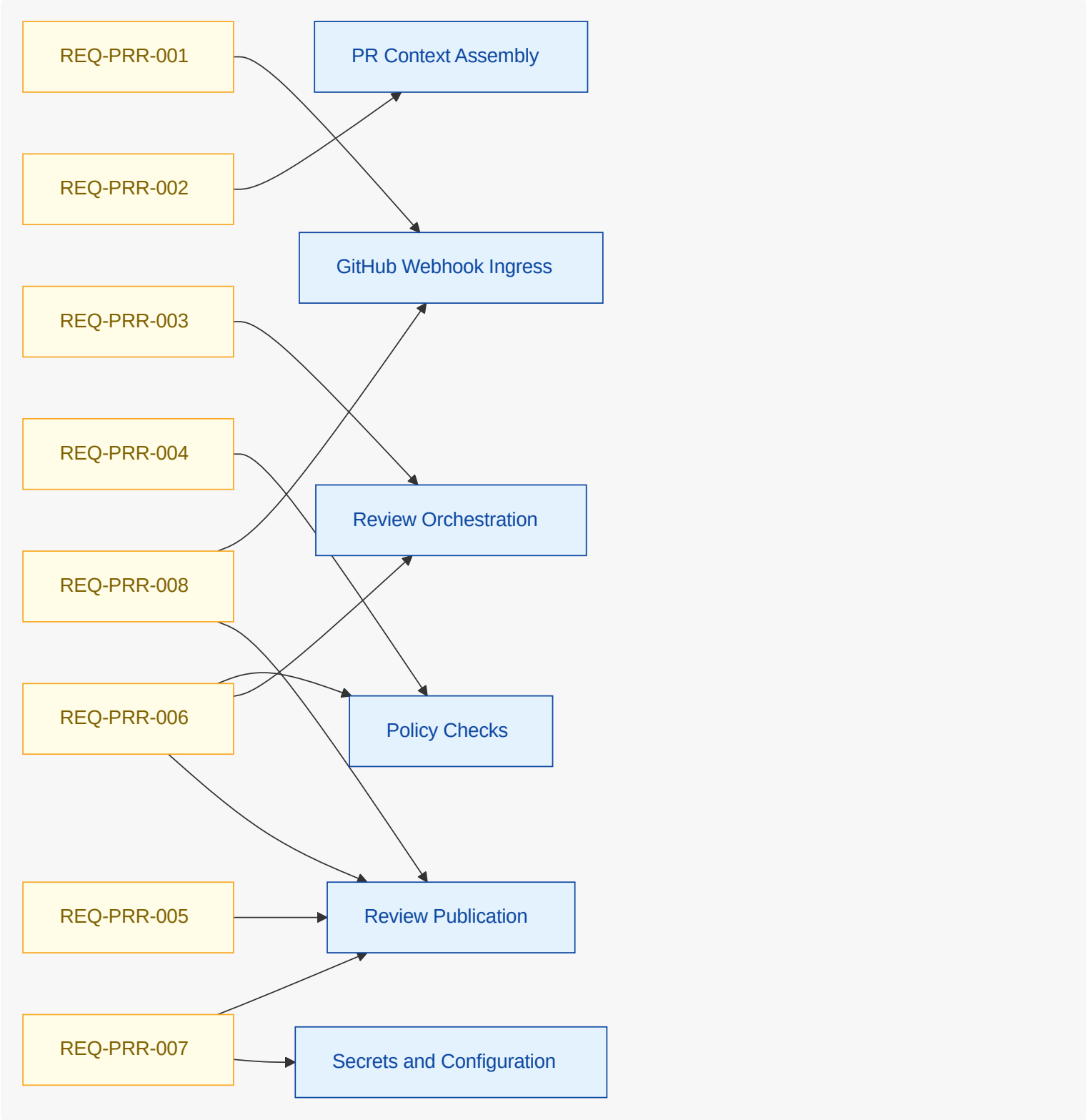
What this diagram shows: authored units and the inferred evidence links that support traceability confidence.

> **NOTE** This graph summarizes inferred ownership/evidence links and is intended for traceability confidence checks.



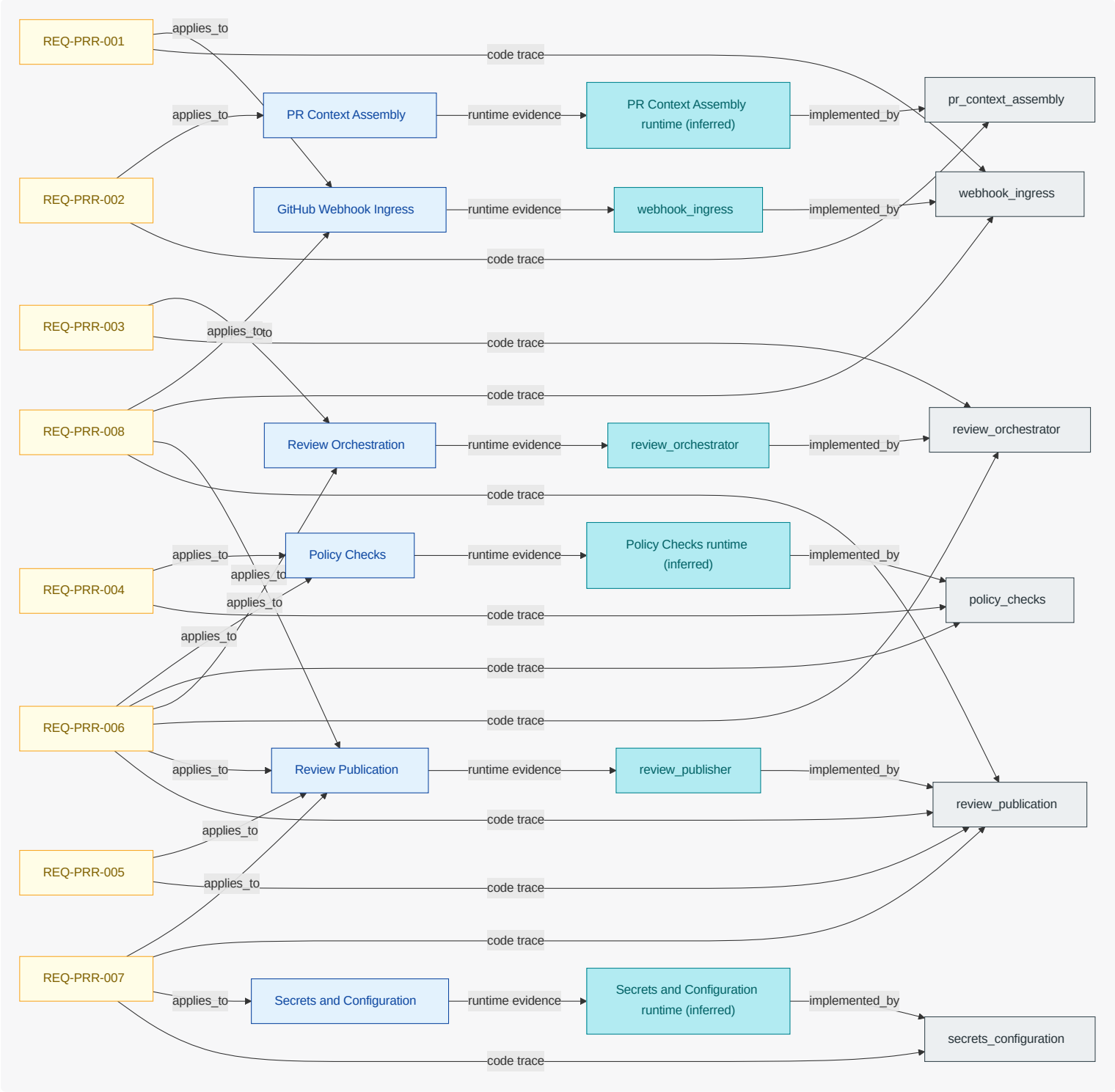
9.5. Requirement-to-Unit Mapping

What this diagram shows: direct authored mapping from requirements to the functional units they apply to.



9.6. Requirement-to-Unit-to-Evidence Coverage

What this diagram shows: requirement-to-unit links extended with inferred runtime/code evidence attached to each unit.



Show requirement-to-unit mappings inferred from appliesTo and authored architecture ownership boundaries.

10. Traceability Appendix

10.1. Requirement Details

10.1.1. REQ-PRR-001

While `pull request review is active`, when `pull request synchronize event is received`, the `pr review system` shall verify `webhook signature` before processing review workflow. Notes: Webhook trust boundary validation.

10.1.2. REQ-PRR-002

While `pull request review is active`, when `pull request opened event is received`, the `pr review system` shall fetch changed files and construct `diff context` for review execution. Notes: Context ingestion from GitHub API.

10.1.3. REQ-PRR-003

Where `ai review is enabled`, when `diff context is prepared`, the `pr review system` shall request `bedrock runtime` analysis for security, architecture, and maintainability findings. Notes: AI-assisted review path.

10.1.4. REQ-PRR-004

Where `policy checks are enabled`, when `pull request review is active`, the `pr review system` shall evaluate deterministic `policy checks` for secret exposure, infrastructure risk, and dependency changes. Notes: Deterministic non-LLM review checks.

10.1.5. REQ-PRR-005

While `pull request review is active`, when `review findings are generated`, the `pr review system` shall publish `check run summary` and inline comments to `repository maintainer`. Notes: Result `publication` contract.

10.1.6. REQ-PRR-006

If `bedrock runtime is unavailable`, then the `pr review system` shall publish deterministic policy-only review outcome. Notes: Graceful degradation `requirement`.

10.1.7. REQ-PRR-007

Where `audit logging is enabled`, when `review findings are published`, the `pr review system` shall persist redacted prompt and `result metadata` for audit traceability. Notes: Evidence and audit lineage.

10.1.8. REQ-PRR-008

If `review request rate limit is exceeded`, then the `pr review system` shall defer processing and notify `repository maintainer`. Notes: Abuse and capacity protection behavior.

10.2. Verification Inventory

What this section shows: verification checks inferred from test artifacts, linked to test code elements and mapped to verified requirements.

10.2.1. Pr Opened Review Flow

Field	Value
ID	`VER-INF-E2E-PR_OPENED_REVIEW_FLOW-6BF0B5`
Kind	e2e
Status	pass
Verifies Requirements	`REQ-PRR-001`, `REQ-PRR-002`, `REQ-PRR-003`, `REQ-PRR-005`
Test Code Elements	tests/e2e/pr_opened_review_flow.yaml

Field	Value
Derived Functional Owners	`FU-GITHUB-WEBHOOK-INGRESS`, `FU-PR-CONTEXT-ASSEMBLY`, `FU-REVIEW-ORCHESTRATION`, `FU-REVIEW-PUBLICATION`
Result Summary	pass:4
Evidence	test-results/contract-checkrun.json, test-results/e2e-pr-opened-review.xml, tests/e2e/pr_opened_review_flow.yaml
Notes	exercises end-to-end pull request opened flow from webhook ingest to published review

10.2.2. Audit Redaction Test

Field	Value
ID	`VER-INF-INTEGRATION-AUDIT_REDACTION_TEST-B108A1`
Kind	integration
Status	pass
Verifies Requirements	`REQ-PRR-007`
Test Code Elements	tests/integration/audit_redaction_test.ts
Derived Functional Owners	`FU-REVIEW-PUBLICATION`, `FU-SECRETS-CONFIGURATION`
Result Summary	pass:1
Evidence	test-results/audit-redaction-report.md, tests/integration/audit_redaction_test.ts
Notes	checks audit evidence redaction for secrets and sensitive payload fields

10.2.3. Policy Only Fallback Test

Field	Value
ID	`VER-INF-INTEGRATION-POLICY_ONLY_FALLBACK_TEST-76E20A`
Kind	integration
Status	pass
Verifies Requirements	`REQ-PRR-004`, `REQ-PRR-006`
Test Code Elements	tests/integration/policy_only_fallback_test.rs
Derived Functional Owners	`FU-POLICY-CHECKS`, `FU-REVIEW-ORCHESTRATION`, `FU-REVIEW-PUBLICATION`
Result Summary	pass:2
Evidence	test-results/integration-policy-fallback.json, tests/integration/policy_only_fallback_test.rs
Notes	verifies policy-only fallback when Bedrock inference is unavailable

10.2.4. Github Check Run Contract Test

Field	Value
ID	`VER-INF-TEST-GITHUB_CHECK_RUN_CONTRACT_TEST-17DBA6`
Kind	test
Status	not-run
Verifies Requirements	`REQ-PRR-005`
Test Code Elements	tests/contract/github_check_run_contract_test.go
Derived Functional Owners	`FU-REVIEW-PUBLICATION`
Result Summary	not-run:1
Evidence	tests/contract/github_check_run_contract_test.go
Notes	validates GitHub check-run contract shape and required review result fields

10.2.5. Rate Limit Deferral Test

Field	Value
ID	`VER-INF-UNIT-RATE_LIMIT_DEFERRAL_TEST-DC9EB2`
Kind	unit
Status	pass
Verifies Requirements	`REQ-PRR-008`
Test Code Elements	tests/unit/rate_limit_deferral_test.go
Derived Functional Owners	`FU-GITHUB-WEBHOOK-INGRESS`, `FU-REVIEW-PUBLICATION`
Result Summary	pass:1
Evidence	test-results/unit-rate-limit.xml, tests/unit/rate_limit_deferral_test.go
Notes	verifies rate-limit deferral behavior for pull request review processing

10.3. Verification Result Mapping

What this section shows: per-requirement result mapping inferred from test-result artifacts.

Check ID	Requirement	Result	Evidence / Notes
`VER-INF-E2E-PR_OPENED_REVIEW_FLOW-6BF0B5`	`REQ-PRR-001`	pass	test-results/e2e-pr-opened-review.xml
`VER-INF-E2E-PR_OPENED_REVIEW_FLOW-6BF0B5`	`REQ-PRR-002`	pass	test-results/e2e-pr-opened-review.xml
`VER-INF-E2E-PR_OPENED_REVIEW_FLOW-6BF0B5`	`REQ-PRR-003`	pass	test-results/e2e-pr-opened-review.xml

Check ID	Requirement	Result	Evidence / Notes
`VER-INF-INTEGRATION-POLICY_ONLY_FALLBACK_TEST-76E20A`	`REQ-PRR-004`	pass	test-results/integration-policy-fallback.json deterministic `policy checks` executed
`VER-INF-E2E-PR_OPENED_REVIEW_FLOW-6BF0B5`	`REQ-PRR-005`	pass	test-results/e2e-pr-opened-review.xml
`VER-INF-TEST-GITHUB_CHECK_RUN_CONTRACT_TEST-17DBA6`	`REQ-PRR-005`	not-run	none No parsed test-result artifact for this `requirement` yet.
`VER-INF-INTEGRATION-POLICY_ONLY_FALLBACK_TEST-76E20A`	`REQ-PRR-006`	pass	test-results/integration-policy-fallback.json policy-only fallback published
`VER-INF-INTEGRATION-AUDIT_REDACTION_TEST-B108A1`	`REQ-PRR-007`	pass	test-results/audit-redaction-report.md
`VER-INF-UNIT-RATE_LIMIT_DEFERRAL_TEST-DC9EB2`	`REQ-PRR-008`	pass	test-results/unit-rate-limit.xml

11. Reference Index

11.1. Authored References

ACT-DEVELOPER

Kind: Actor Name: Developer

Opens pull requests and receives review feedback.

ACT-PLATFORM-OPERATOR

Kind: Actor Name: Platform Operator

Operates Lambda deployment and runtime controls.

ACT-REPOSITORY-MAINTAINER

Kind: Actor Name: Repository Maintainer

Owns merge decisions and repository-level quality gates.

ACT-SECURITY-ENGINEER

Kind: Actor Name: Security Engineer

Curates policy checks and triages security findings.

AV-PR-SPAM-ABUSE

Kind: Attack Vector Name: Pull Request Spam Abuse

High-rate PR event flooding that can exhaust review capacity.

AV-PROMPT-INJECTION-IN-DIFF

Kind: Attack Vector Name: Prompt Injection in Diff

Malicious prompt-like content embedded in pull request diffs.

AV-SECRET-LEAKAGE-IN-REVIEW

Kind: Attack Vector Name: Secret Leakage in Review Output

Generated findings accidentally exposing secrets/tokens in comments.

AV-SPOOFED-GITHUB-WEBHOOK

Kind: Attack Vector Name: Spoofed GitHub Webhook

Forged webhook payload attempting unauthorized review execution.

FG-PLATFORM-OPERATIONS

Kind: Functional Group Name: Platform Operations

Lambda deployment lifecycle, runtime operations, and secrets/configuration.

FG-PR-INTEGRATION

Kind: Functional Group Name: PR Integration

Ingress and context assembly for GitHub pull request events.

FG-REVIEW-INTELLIGENCE

Kind: Functional Group Name: Review Intelligence

AI and deterministic review analysis with result publication.

FU-GITHUB-WEBHOOK-INGRESS

Kind: Functional Unit Name: GitHub Webhook Ingress

Verifies webhook signatures, validates event type, and starts review workflow routing.

FU-LAMBDA-RUNTIME-OPERATIONS

Kind: Functional Unit Name: Lambda Runtime Operations

Manages Lambda packaging/deployment lifecycle and operational health guardrails.

FU-POLICY-CHECKS

Kind: Functional Unit Name: Policy Checks

Runs deterministic checks for secrets, risky IaC changes, and repository policy constraints.

FU-PR-CONTEXT-ASSEMBLY

Kind: Functional Unit Name: PR Context Assembly

Fetches changed files and diff metadata and prepares bounded review context payloads.

FU-REVIEW-ORCHESTRATION

Kind: Functional Unit Name: Review Orchestration

Calls Bedrock models, merges AI and deterministic findings, and handles graceful degradation paths.

FU-REVIEW-PUBLICATION

Kind: Functional Unit Name: Review Publication

Publishes check-run summaries and inline comments back to GitHub pull requests.

FU-SECRETS-CONFIGURATION

Kind: Functional Unit Name: Secrets and Configuration

Manages GitHub App credentials, Bedrock config, and redaction-safe runtime configuration.

REF-AWS-BEDROCK-RUNTIME

Kind: Referenced Element Name: AWS Bedrock Runtime API

external_service_endpoint / runtime

REF-AWS-SDK-GO

Kind: Referenced Element Name: AWS SDK for Go

third_party_library / code

REF-AWS-SECRETS-MANAGER

Kind: Referenced Element Name: AWS Secrets Manager

platform_service / runtime

REF-GITHUB-APP-API

Kind: Referenced Element Name: GitHub App API

external_service_endpoint / runtime

REF-GITHUB-REST-SDK

Kind: Referenced Element Name: GitHub REST SDK

third_party_library / code

REF-POLICY-RULESET

Kind: Referenced Element Name: Review Policy Ruleset

policy_bundle / code

REQ-PRR-001

Kind: Requirement Name: REQ-PRR-001 Resolves to: `canonical` While pull request review is active, when pull request synchronize event is received, the pr review system shall verify webhook signature before processing review workflow.

REQ-PRR-002

Kind: Requirement Name: REQ-PRR-002 Resolves to: `canonical` While pull request review is active, when pull request opened event is received, the pr review system shall fetch changed files and construct diff context for review execution.

REQ-PRR-003

Kind: Requirement Name: REQ-PRR-003 Resolves to: `canonical` Where ai review is enabled, when diff context is prepared, the pr review system shall request bedrock runtime analysis for security, architecture, and maintainability findings.

REQ-PRR-004

Kind: Requirement Name: REQ-PRR-004 Resolves to: `canonical` Where policy checks are enabled, when pull request review is active, the pr review system shall evaluate deterministic policy checks for secret exposure, infrastructure risk, and dependency changes.

REQ-PRR-005

Kind: Requirement Name: REQ-PRR-005 Resolves to: `canonical` While pull request review is active, when review findings are generated, the pr review system shall publish check run summary and inline comments to repository maintainer.

REQ-PRR-006

Kind: Requirement Name: REQ-PRR-006 Resolves to: `canonical` If bedrock runtime is unavailable, then the pr review system shall publish deterministic policy-only review outcome.

REQ-PRR-007

Kind: Requirement Name: REQ-PRR-007 Resolves to: `canonical` Where audit logging is enabled, when review findings are published, the pr review system shall persist redacted prompt and result metadata for audit traceability.

REQ-PRR-008

Kind: Requirement Name: REQ-PRR-008 Resolves to: `canonical` If review request rate limit is exceeded, then the pr review system shall defer processing and notify repository maintainer.

11.2. Catalog References

developer

Engineer opening and updating pull requests. Canonical: `open`

platform operator

Owner of deployment/runtime operations and reliability controls. Canonical: `open`

repository maintainer

Owner of merge decisions and repository quality gates. Canonical: `open`

security engineer

Owner of security policy checks and finding triage. Canonical: `open`

ai review is enabled

Feature flag enabling model-assisted review. Canonical: `open`

audit logging is enabled

Feature flag enabling audit metadata persistence. Canonical: `open`

aws sdk for go

SDK dependency used for AWS runtime integrations. Canonical: `open`

bedrock runtime

External Bedrock runtime endpoint used for model inference. Canonical: `open`

bedrock runtime is unavailable

Event raised when Bedrock inference path is degraded. Canonical: `open`

pr context assembly

Fetches changed files and prepares bounded review context payloads. Canonical: `open`

diff context is prepared

Event indicating context assembly completed and is ready for analysis. Canonical: `open`

policy checks

Evaluates deterministic repository policy checks. Canonical: `open`

review findings are generated

Event indicating analysis output is ready for publication. Canonical: `open`

review findings are published

Event indicating review output has been published to GitHub checks/comments. Canonical: `open`

github app api

External GitHub API used for pull request metadata and review publication. Canonical: `open`

pr integration

Functional group for webhook ingress and pull request context intake. Canonical: `open`

github rest sdk

SDK dependency used for GitHub App REST interactions. Canonical: `open`

lambda runtime operations

Manages runtime deployment and operational controls. Canonical: `open`

repository maintainer

Owner of merge decisions and repository quality gates. Canonical: `open`

normal mode is active

Default operating mode using AI plus deterministic checks. Canonical: `open`

review orchestration

Coordinates Bedrock analysis with deterministic checks and fallback logic. Canonical: `open`

platform operator

Owner of deployment/runtime operations and reliability controls. Canonical: `open`

policy ruleset

Deterministic review policy bundle used during analysis. Canonical: `open`

policy checks are enabled

Feature flag enabling deterministic policy checks. Canonical: `open`

developer

Engineer opening and updating pull requests. Canonical: `open`

pull request opened event is received

Event raised when a new pull request is opened. Canonical: `open`

review intelligence

Functional group for AI and deterministic review analysis and publication. Canonical: `open`

pull request spam abuse

High-volume event abuse causing review capacity exhaustion. Canonical: `open`

pull request synchronize event is received

Event raised when new commits are pushed to an open pull request. Canonical: `open`

prompt injection in diff

Malicious prompt-like content embedded in pull request code changes. Canonical: `open`

review publication

Publishes review findings to GitHub checks and comments. Canonical: `open`

review request rate limit is exceeded

Event raised when PR review rate guardrails are hit. Canonical: `open`

pull request review is active

State where review processing is currently active for a pull request. Canonical: `open`

platform operations

Functional group for deployment lifecycle and secret/config controls. Canonical: `open`

secrets and configuration

Manages credentials and secure runtime configuration. Canonical: `open`

secret leakage in review output

Unintended credential disclosure in generated review findings. Canonical: `open`

aws secrets manager

Managed secret store for application credentials and config values. Canonical: `open`

security engineer

Owner of security policy checks and finding triage. Canonical: `open`

webhook signature is valid

Condition indicating webhook HMAC verification succeeded. Canonical: `open`

github webhook ingress

Validates webhook signatures and routes trusted pull request events. Canonical: `open`

spoofed github webhook

Forged webhook request attempting unauthorized review execution. Canonical: `open`

pull request spam abuse

High-volume event abuse causing review capacity exhaustion. Canonical: `open`

prompt injection in diff

Malicious prompt-like content embedded in pull request code changes. Canonical: `open`

secret leakage in review output

Unintended credential disclosure in generated review findings. Canonical: `open`

spoofed github webhook

Forged webhook request attempting unauthorized review execution. Canonical: `open`

webhook signature is valid

Condition indicating webhook HMAC verification succeeded. Canonical: `open`

check run summary

Aggregated review result posted as a GitHub check run. Canonical: `open`

diff context

Pull request file changes and metadata prepared for analysis. Canonical: `open`

redacted audit metadata

Sanitized prompt and finding metadata retained for audit. Canonical: `open`

result metadata

Metadata describing review output characteristics for audit and reporting. Canonical: `open`

webhook signature

Signature used to verify webhook authenticity. Canonical: `open`

bedrock runtime is unavailable

Event raised when Bedrock inference path is degraded. Canonical: `open`

diff context is prepared

Event indicating context assembly completed and is ready for analysis. Canonical: `open`

review findings are generated

Event indicating analysis output is ready for publication. Canonical: `open`

review findings are published

Event indicating review output has been published to GitHub checks/comments. Canonical: `open`

pull request opened event is received

Event raised when a new pull request is opened. Canonical: `open`

pull request synchronize event is received

Event raised when new commits are pushed to an open pull request. Canonical: `open`

review request rate limit is exceeded

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Feature flag enabling model-assisted review. Canonical: `open`

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Feature flag enabling deterministic policy checks. Canonical: `open`

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Functional group for deployment lifecycle and secret/config controls. Canonical: `open`

pr integration

Functional group for webhook ingress and pull request context intake. Canonical: `open`

review intelligence

Functional group for AI and deterministic review analysis and publication. Canonical: `open`

github webhook ingress

Validates webhook signatures and routes trusted pull request events. Canonical: `open`

lambda runtime operations

Manages runtime deployment and operational controls. Canonical: `open`

policy checks

Evaluates deterministic repository policy checks. Canonical: `open`

pr context assembly

Fetches changed files and prepares bounded review context payloads. Canonical: `open`

review orchestration

Coordinates Bedrock analysis with deterministic checks and fallback logic. Canonical: `open`

review publication

Publishes review findings to GitHub checks and comments. Canonical: `open`

secrets and configuration

Manages credentials and secure runtime configuration. Canonical: `open`

normal mode is active

Default operating mode using AI plus deterministic checks. Canonical: `open`

bedrock runtime

External Bedrock runtime endpoint used for model inference. Canonical: `open`

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SDK dependency used for GitHub App REST interactions. Canonical: `open`

policy ruleset

Deterministic review policy bundle used during analysis. Canonical: `open`

pull request review is active

State where review processing is currently active for a pull request. Canonical: `open`

actor

A person or operational role that interacts with functional units. Canonical: `open`

attack vector

A technical misuse or attack path that targets functional, referenced, or runtime elements. Canonical: `open`

authored mapping

An explicit relationship declared in architecture inputs between authored or referenced elements. Canonical: `open`

code element

An inferred code structure or ownership element discovered from source trees and build metadata. Canonical: `open`

functional group

A major authored capability area that groups related functional units. Canonical: `open`

functional unit

An authored working unit inside a functional group that owns specific behavior. Canonical: `open`

inferred mapping

A discovered relationship that links inferred runtime/code elements upward to authored design. Canonical: `open`

referenced element

An architecture-relevant external, platform, or third-party dependency represented by role. Canonical: `open`

requirement

A structured requirement statement that defines expected system behavior and maps to functional ownership. Canonical: `open`

runtime element

An inferred runtime realization element discovered from infrastructure and deployment sources. Canonical: `open`

upward linking

Rule where runtime and code elements point to stable functional groups/units; authored architecture does not depend on inferred IDs. Canonical: `open`

verification check

An inferred or authored verification artifact that validates requirement behavior with test evidence. Canonical: `open`

11.3. Inferred Runtime References

ID	Kind	Owner	Description	Source
review_orchestrator	Runtime lambda_function	FU-REVIEW-ORCHESTRATION	assembles diff context and orchestrates deterministic plus Bedrock-backed review analysis	examples/bedrock-pr-review-github-app-sample/infra/terraform/main.tf
review_publisher	Runtime lambda_function	FU-REVIEW-PUBLICATION	publishes check run summaries and inline review comments back to GitHub	examples/bedrock-pr-review-github-app-sample/infra/terraform/main.tf
webhook_ingress	Runtime lambda_function	FU-GITHUB-WEBHOOK-INGRESS	validates GitHub webhook signature and normalizes pull request event payloads	examples/bedrock-pr-review-github-app-sample/infra/terraform/main.tf

11.4. Inferred Code References

ID	Kind	Owner	Description	Source
aws-sdk-go-v2/secretsmanager	Code library_external	FU-SECRETS-CONFIGURATION	n/a	secrets_configuration.ts
github-rest-sdk	Code library_external	FU-PR-CONTEXT-ASSEMBLY	n/a	pr_context_assembly.ts
github-rest-sdk/publication-client	Code library_external	FU-REVIEW-PUBLICATION	n/a	review_publication.ts
serde_json::json	Code library_external	FU-REVIEW-ORCHESTRATION	n/a	review_orchestrator.rs
crypto/hmac	Code library_stdlib	FU-GITHUB-WEBHOOK-INGRESS	n/a	webhook_ingress.go
crypto/sha256	Code library_stdlib	FU-GITHUB-WEBHOOK-INGRESS	n/a	webhook_ingress.go
encoding/hex	Code library_stdlib	FU-GITHUB-WEBHOOK-INGRESS	n/a	webhook_ingress.go
fmt	Code library_stdlib	FU-GITHUB-WEBHOOK-INGRESS	n/a	webhook_ingress.go
log	Code library_stdlib	FU-LAMBDA-RUNTIME-OPERATIONS	n/a	lambda_runtime_ops.go
lambda_runtime_ops.go	Code source_file	FU-LAMBDA-RUNTIME-OPERATIONS	wires Lambda runtime entrypoints and shared execution dependencies	lambda_runtime_ops.go

ID	Kind	Owner	Description	Source
policy_checks.rs	Code source_file	FU-POLICY-CHECKS	evaluates policy and style rules before review publication	policy_checks.rs
pr_context_assembly.ts	Code source_file	FU-PR-CONTEXT-ASSEMBLY	collects pull request diff context and prepares review input payloads	pr_context_assembly.ts
review_orchestrator.rs	Code source_file	FU-REVIEW-ORCHESTRATION	coordinates deterministic and AI review steps for pull requests	review_orchestrator.rs
review_publication.ts	Code source_file	FU-REVIEW-PUBLICATION	publishes review comments and check-run outcomes back to GitHub	review_publication.ts
secrets_configuration.ts	Code source_file	FU-SECRETS-CONFIGURATION	loads GitHub and Bedrock secret configuration for review execution	secrets_configuration.ts
webhook_ingress.go	Code source_file	FU-GITHUB-WEBHOOK-INGRESS	validates GitHub webhook signatures and routes pull request events	webhook_ingress.go
CODE-BUILDDIFFCONTEXT	Code symbol	FU-PR-CONTEXT-ASSEMBLY	n/a	pr_context_assembly.ts:15
CODE-FALLBACKPOLICYONLY	Code symbol	FU-REVIEW-ORCHESTRATION	n/a	review_orchestrator.rs:19
CODE-FETCHCHANGEDFILES	Code symbol	FU-PR-CONTEXT-ASSEMBLY	n/a	pr_context_assembly.ts:9
CODE-GETGITHUBAPPPRIVATEKEY	Code symbol	FU-SECRETS-CONFIGURATION	n/a	secrets_configuration.ts:9
CODE-NOTIFYMAINTAINERDEFERRAL	Code symbol	FU-REVIEW-PUBLICATION	n/a	review_publication.ts:27
CODE-POLICYONLYSUMMARY	Code symbol	FU-POLICY-CHECKS	n/a	policy_checks.rs:21
CODE-PUBLISHCHECKRUN	Code symbol	FU-REVIEW-PUBLICATION	n/a	review_publication.ts:9
CODE-RECORDREDACTEDAUDITMETADATA	Code symbol	FU-REVIEW-PUBLICATION	n/a	review_publication.ts:22
CODE-REQUESTBEDROCKREVIEW	Code symbol	FU-REVIEW-ORCHESTRATION	n/a	review_orchestrator.rs:9
CODE-ROUTEPULLREQUESTEVENT	Code symbol	FU-GITHUB-WEBHOOK-INGRESS	n/a	webhook_ingress.go:25
CODE-RUNDETERMINISTICCHECKS	Code symbol	FU-POLICY-CHECKS	n/a	policy_checks.rs:7

ID	Kind	Owner	Description	Source
CODE-VERIFYSIGNATURE	Code symbol	FU-GITHUB-WEBHOOK-INGRESS	n/a	webhook_ingress.go:15

11.5. Verification References

ID	Kind	Status	Description	Source	Canonical
VER-INF-E2E-PR_OPENED_REVIEW_FLOW-6BF0B5	e2e	pass	exercises end-to-end pull request opened flow from webhook ingest to published review	test-results/contract-checkrun.json (+2)	canonical
VER-INF-INTEGRATION-AUDIT_REDACTION_TEST-B108A1	integration	pass	checks audit evidence redaction for secrets and sensitive payload fields	test-results/audit-redaction-report.md (+1)	canonical
VER-INF-INTEGRATION-POLICY_ONLY_FALLBACK_TEST-76E20A	integration	pass	verifies policy-only fallback when Bedrock inference is unavailable	test-results/integration-policy-fallback.json (+1)	canonical
VER-INF-TEST-GITHUB_CHECK_RUN_CONTRACT_TEST-17DBA6	test	not-run	validates GitHub check-run contract shape and required review result fields	tests/contract/github_check_run_contract_test.go	canonical
VER-INF-UNIT-RATE_LIMIT_DEFERRAL_TEST-DC9EB2	unit	pass	verifies rate-limit deferral behavior for pull request review processing	test-results/unit-rate-limit.xml (+1)	canonical